

Cyber Security and Forensics

AICT014-4-2-CSF

Version-VE3

Introduction and Overview

Lecturer Information

- Lecturer Name: Zaireeda Mohd Fauzee
- Email: zaireeda@apu.edu.my
- Communication – MS. Teams (On channel)

Pre-requisites for this module

- None

Synopsis

- This module is an **introduction to a range of technologies** associated with **computer security** and **digital forensics**.
- It will expose students to **tools, concepts and mechanisms** in achieving an ideal information security environment.
- Some of the content is mapped to CompTIA Security+ certification to provide knowledge on industry-relevant technologies and practices.
- As part of digital forensics exposure, students will also be guided to **perform data tracing and evidence gathering** to simulate a part of forensics investigation following the correct methodology and procedures.

Course Learning Outcomes, CLOs

1. Discuss concepts and implementation of security and forensic techniques and technologies in different scenarios.
2. Identify the existence of vulnerabilities in the given virtual machine using appropriate tools.
3. Perform memory acquisition on a given virtual machine to create forensic images for investigation.

CLO - Course Learning Outcomes

PLO - Programme Learning Outcomes

Mapping of CLOs with MOEs Domain

Course Learning Outcomes	Programme Learning Outcomes (PLO)											Teaching Methods	Assessment Methods
	Knowledge and Understanding	Cognitive Skills	Practical Skills	Interpersonal Skills	Communication Skills	Digital Skills	Numeracy Skills	Personal Skills	Entrepreneurial Skills	Ethics and professionalism			
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10			
CLO1	v											Lecture	Final Exam
CLO2								v				Tutorial, Case Study	Individual Assignment - Vulnerability Scanning
CLO3						v						Tutorial, Case Study	Individual Assignment - Memory Acquisition

PLO1 - Knowledge and Understanding
 PLO6 - Digital Skills
 PLO8 – Personal Skills

Teaching Strategies

- Lecture
- Tutorial
- Case Study (Individual and Group)

Assessment Methods

- Final Exam (50%): CLO1
- Group Assignment & Individual, Presentation (50%): CLO2 & CLO3
 - Appraise Security issues
 - Propose solutions
 - Presentation

Student Learning Time (SLT)

- **Course Credit Value: 4**
- **Total Learning Hours:**
 - Lecture: 28 hours per semester
 - Tutorial / Case Study : 28 hours per semester

Module Content Outline

Lecture

Computer Ethics & Security Concepts

Cybersecurity Threats

Vulnerability Assessment & Penetration Testing

Cryptography & PKI

Access Control

Securing Network, Data & Applications

Security Policies, Standards, and Compliance

Risk Management and Privacy

Digital Forensics

Tutorial / Case Study

Case Study: Installing and Configuring Vulnerability Scanner

Case Study: Vulnerability Scanning

Forensic Tools & Techniques

Case Study: RAM & HDD Imaging

Case Study: RAM & HDD Analysis

Expectations

1. Abide by ALL rules and regulations of APU.
2. Proper attire.
3. No speaking of dialects.
4. Attendance is compulsory. Valid Medical Certs must be supported in any absence from class.
5. **Three cases of Late will be equal to 1 absence.**
6. Use **proper academic references – APA Referencing only.**
7. Academic Dishonesty / Plagiarism is a serious offence. Any suspicions will be referred to the University's Academic Dishonesty Board.
8. Formal assessments must be submitted on time in the specified format given. Failure to meet deadlines will be treated as non-submission and no marks will be awarded. Incomplete submissions will be subjected to penalty of mark deductions or forfeit.

Achievement Requirements: Diploma Programmes

Undergraduate:

Marks	Alphabetical Grade	Grading Point	Classification
80-100	A+	4.0	Distinction
75-79	A	3.7	
70-74	B+	3.3	Credit
65-69	B	3.0	
60-64	C+	2.7	Pass
55-59	C	2.3	
50-54	C-	2.0	
40-49	D	1.7	Fail (marginal)
30-39	F+	1.3	Fail
20-29	F	1.0	Fail
0-19	F-	0	Fail

What support is available for you

1. Consultation hours (Refer to the i-consult system)

2. Resources

a. Tutorials Materials

- MS Teams announcement
- Moodle

b. Reference material

c. Essential Reading

- Chapple, M., Seidl, D. (2021). CompTIA Security+ Study Guide: Exam SY0-601. 8th ed. John Wiley & Sons. USA. ISBN: 978-1119736257
- Nelson, B., Philips, A. and Steuart, C. (2019) Guide To Computer Forensics and Investigations - Standalone Book. 6th ed. Cengage. USA. ISBN: 978-1337568944
- Introduction to security and network forensics Technologies, William J. Buchanan 2011, CRC Press ISBN 9780849335686

d. Internet resources

Tutorial Activities

- Windows-based host machine
- Virtual Machines will be provided
- VMware workstation player / professional

What we will cover next

Week 1

- Tutorial:
 - Virtual Machines Installation